MIL-W-283H
25 June 1987
SUPERSEDING
MIL-W-283G
18 December 1979

MILITARY SPECIFICATION

WATER PURIFICATION TABLET, IODINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and grade of iodine water purification tablet suitable for purification of water in canteens (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-186	- Container, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6850

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-140 - Tablet (For Medicinal Purposes)

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Federal Insecticide, Fungicide, and Rodenticide Act as Amended and Regulations Promulgated Thereunder (40 CFR Part 162)

(Application for copies should be addressed to the Environmental Protection Agency, 401 M Street, S.W. Washington, DC 20460.)

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D5 - Penetration of Bituminous Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.

3.1.1 Tetraglycine hydroperiodide. Tetraglycine hydroperiodide shall conform to the requirements specified in table I when tested as specified in 4.2.1.1.

TABLE I. Requirements for tetraglycine hydroperiodide

Ingredient	Minimum percent	Maximum percent
Active iodine	39.50	42.60
Ash	-	0.50
Total nitrogen	7.35	7.85
Titratable iodine after 7 days storage in a tightly sealed bottle at 130°F	34.50	-

* 3.1.2 Sodium acid pyrophosphate, anhydrous. Sodium acid pyrophosphate, anhydrous ($Na_2H_2P_2O_7$) shall meet the requirements of the Food Chemicals Codex grade.

3.1.3 Talc. The inert material shall consist of non-asbestiform talc, U.S.P. suitable for its intended use. It shall be selected to provide good tableting properties without affecting the solubility, friability, and stability of the tablets beyond the specified limits. In addition, the talc shall lose not more than 0.3 percent of its weight (moisture) when tested as specified in 4.2.1.1.

* 3.2 Form and composition. The tablets shall be uncoated, be of a firm double convex form, contain no added coloring material, and shall be of the composition specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

TABLE II. Tablet composition

Ingredient	Minimum milligrams	Maximum milligrams
Tetraglycine hydroperiodide	19.3	21.3
Sodium acid pyrophosphate, anhydrous	82.5	92.3
Talc, U.S.P.	-	6.0

3.3 Physical and chemical requirements.

3.3.1 Weight. The average tablet weight shall not exceed 120 milligrams when tested as specified in 4.2.4. The individual tablet weight deviations shall conform to the tolerances for tablet weights as outlined in the Weight Variation Test for Tablets referenced in the United States Pharmacopeia.

3.3.2 Loss on drying. Tablets shall have a maximum loss on drying of 0.20 percent when tested as specified in 4.2.4, regardless of the maximum allowable loss on drying of the individual components of the tablet.

3.3.3 Titratable iodine. Tablets shall yield not less than 7.60 milligrams nor more than 9.00 milligrams per tablet of titratable iodine when tested as specified in 4.2.4.

3.3.4 Solubility. Tablets, when tested as specified in 4.2.4, shall dissolve in not more than 200 seconds. If the tablets do not dissolve within the specified time, they will be acceptable providing the titratable iodine in solution at the end of 200 seconds when divided by 8 is not less than 7.60 mg when tested in accordance with 4.2.4.

3.3.5 Friability. The tablets shall not lose more than 10.0 percent in weight when tested as specified in 4.2.4.

3.3.6 Stability.

3.3.6.1 Loss of iodine after storage. Tablets as packaged in the sealed, glass bottle after 7 days storage at 130°F shall show a loss of not more than 5.0 percent of the original titratable iodine content when tested as specified in 4.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

3.3.6.2 Loss of solubility after storage. Tablets as packaged in the sealed, glass bottle after 7 days storage at 130°F, when tested as specified in 4.2.4, shall dissolve in not more than 300 seconds. If the tablets do not dissolve within the 300 seconds time limit, they will be acceptable providing the titratable iodine in the 300 mL aliquot at the end of 300 seconds when divided by 8 is not less than 95.0 percent of the original titratable iodine.

3.4 Pledget material. Pledget material shall conform to the following requirements prior to use in packaging.

3.4.1 Glass fiber pledget. Glass fiber pledget material for the void space in immediate container shall consist of long fibers of calcium borosilicate glass. It shall be white in color and free from dust, grease, and other foreign contaminants.

3.4.1.1 Glass fiber diameter. The average diameter of the fibers shall be between 0.00010 and 0.00036 inch, and the individual fibers shall not deviate from the average by more than ± 50 percent when tested as specified in 4.2.1.1.

3.4.1.2 Moisture content. Moisture content of the glass fibers shall be not greater than 0.5 percent when tested as specified in 4.2.1.1.

3.4.1.3 Loss on ignition. The loss of weight of the glass fibers on ignition shall be not greater than 1.0 percent when tested as specified in 4.2.1.1.

* 3.5 Labeling. The labeling shall be registered under the Federal Insecticide, Fungicide, and Rodenticide Act and shall be in compliance with its requirements. The contractor shall use EPA Registration No. 40510-3 assigned to the U.S. Army Natick Research, Development, and Engineering Center (GL), Natick, MA 01760-5011, and not the EPA Registration assigned for a commercial version of the document item. The contractor shall use the EPA Establishment number assigned to his manufacturing plant. The EPA approved label shall be as shown in figure 1.

* 3.5.1 Labeling of bottles. The label shall be of paper in one color, printed with nonsmearing black ink, and shall be firmly affixed to the bottle with an adhesive so that it shall withstand soaking in water at 70°F for 24 hours without loosening or rendering the printed matter illegible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

* 3.5.2 Instruction sheet (twelve-tablet bottle only). Twelve instruction sheets, containing the information specified in figure 1, shall be furnished in each intermediate pack of 12-tablet bottles (see 5.1.1.2.1). Each instruction sheet shall be printed with a black nonsmearing ink and shall be of such size and print to be easily read. The paper used shall be bleached, high wet strength kraft paper having a minimum basis weight of 27 pounds per ream (24 by 36 - 500).

* 3.6 Workmanship. The tablets shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

4.2.1.1 Component testing. In addition to any testing required in referenced documents, components shall be tested for characteristics as specified in table III. The lot shall be expressed in pounds of chemical components, pledget material (glass fiber), or wax, as applicable. The sample for testing the chemical components and wax shall be a 1/4 pound composite obtained by combining equal portions selected at random throughout the lot. The number of portions of products to make up the composite shall be as indicated below. The composite samples shall be thoroughly mixed, placed in airtight containers, sealed, and marked with the identifying lot number and date. The sample unit for the pledget material shall be a 1/4 pound sample. The number of units of pledget material shall be as indicated below. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the composite or sample unit, as applicable, fails to meet any test requirement specified.

Lot size (pounds)	Sample size (sample unit)
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Each lot of tetraglycine hydroperiodide, anhydrous sodium acid pyrophosphate, and talc shall be tested to determine compliance with the requirements specified herein.

TABLE III. Component tests

Characteristic	Requirement para	Test method	No. of determinations	Results reported to nearest
Tetraglycine hydroperiodide:				
Active iodine	3.1.1	4.3.1	2	0.01 percent
Ash	3.1.1	4.3.2	2	0.01 percent
Total nitrogen	3.1.1	4.3.3	2	0.01 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

TABLE III. Component tests - Continued

Characteristic	Requirement para	Test method	No. of determinations	Results reported to nearest
Tetraglycine hydroperiodide: - continued				
Titratable iodine after 7 days storage in a tightly sealed bottle at 130°F	3.1.1	4.3.1	2	0.01 percent
Sodium acid pyrophosphate, anhydrous	3.1.2	1/	-	-
Talc:				
Characteristics (U.S.P.)	3.1.3	1/	-	-
Moisture content	3.1.3	4.3.4	2	0.01 percent
Pledget material:				
Calcium borosilicate glass	3.4.1	1/	-	-
Diameter of fibers	3.4.1.1	4.3.5	3	0.00001 inch
Moisture content	3.4.1.2	4.3.4	2	0.01 percent
Loss on ignition	3.4.1.3	4.3.6	2	0.01 percent
Closure (additional protection):				
Vinyl film facing	5.1.1.1.3	1/	-	-
Wax coating (optional):				
Composition	5.1.1.1.3	1/	-	-
Melting point	5.1.1.1.3	1/	-	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

TABLE III. Component tests - Continued

Characteristic	Requirement para	Test method	No. of determinations	Results reported to nearest
Wax (secondary closure):				
Composition	5.1.1.1.3	1/	-	-
Melting point	5.1.1.1.3	1/	-	-
Needle penetration	5.1.1.1.3	ASTM D5	4	Whole number

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.2.2 In-process inspection. In-process inspection shall be in accordance with FED-STD-140.

4.2.3 End item examination. End item examination shall be in accordance with FED-STD-140.

4.2.4 End item testing. The end item shall be tested for the characteristics specified in table IV. For purposes of sampling, the lot shall be expressed in units of bottles. The sample unit for testing shall be five sealed unopened bottles randomly selected from different shipping containers throughout the lot. The sample size shall be as indicated below. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fails to meet any test requirement specified.

Lot size (bottles)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

TABLE IV. End item tests

Characteristic	Requirement para	Test method	No. of determinations	Results reported to nearest
Composition	3.2	1/	-	-
Weight	3.3.1	4.3.7	1	1 mg
Loss on drying	3.3.2	4.3.8	1	0.01 percent
Titratable iodine	3.3.3	4.3.1	1	0.01 mg
Solubility	3.3.4	4.3.9.1	1	1 second 2/
Friability	3.3.5	4.3.10	1	0.1 percent
Loss of iodine after storage	3.3.6.1	4.3.1	1	0.1 percent
Loss of solubility after storage	3.3.6.2	4.3.9.2	1	1 second 2/
Wax coating stability	5.1.1.1.3	4.3.11	1	Pass or fail

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ If the tablets do not meet the dissolving time requirement, report titratable iodine to nearest 0.01 mg.

* 4.2.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of intermediate packs per shipping container is more or less than specified Number of unit packs per intermediate container is less than specified
Instruction sheet (twelve tablet bottle only)	Less than twelve instruction sheets in each intermediate pack Not placed as specified Incomplete, incorrect, or illegible

* 4.2.5.1 Unit pack examination. Unit packs shall be examined for the defects listed in table V. The lot size shall be expressed in units of filled unit packs. The sample shall be one filled unit pack. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE V. Unit pack defects

Examine	Defect	Classification	
		Major	Minor
Assembly	Not unit packed in a glass bottle fitted with a metal screw cap, or a plastic screw cap, with a secondary closure of wax coating	101	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

TABLE V. Unit pack defects - Continued

Examine	Defect	Classification	
		Major	Minor
Bottle	Not amber-colored glass	102	
	Not clean	103	
	Broken, cracked, or chipped	104	
	Not square or round		201
	Not completely dry	105	
Dimensions of bottle	Varies from the required dimensions including tolerances	106	
Cap closure	Not threaded, knurled metal, rolled edge screw cap, or dark plastic screw cap	107	
	Not completely and securely screwed into position to form a tight seal	108	
	Not completely dry	109	
	Not coated to resist iodine corrosion	110	
	Not clean; any evidence of rust	111	
	Cap not plastic, verticle ribs missing (twelve-tablet bottle only)	112	
Cap insert	Insert missing; distorted, rusty, or not clean	113	
	Insert pad not faced with vinyl film	114	
	Insert pad does not completely cover entire inner top of cap, or loose so that it falls out of cap	115	
Pledget material	Head space between tablets and cap closure not completely filled with pledget material (fiberglass)	116	
	Not clean, not free from dust or grease	117	
	Any foreign contaminants or glass pellets in pledget	118	
	Not white, clean, and odorless		202
	Not completely dry	119	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

TABLE V. Unit pack defects - Continued

Examine	Defect	Classification	
		Major	Minor
Secondary wax closure	Missing; cracked or loose	120	
	Does not effect a complete and neat seal of cap and bottle juncture	121	
	Does not completely cover lower edge of cap and bottle shoulder	122	
	Not clear and homogeneous		203
	Presence of air bubbles or voids between lower edge of cap and bottle shoulder	123	
	Obliterates label markings	124	
Labeling on bottles	Label or printing pertaining to the applicable information and instructions for end item use, missing, illegible, incomplete, or smears	125	
	Not firmly affixed to bottle; printing not with nonsmearing black ink	126	
	Label not of paper in one color		204
	EPA registration number not as specified	128	

4.2.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

Examine	Defect
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.3 Methods of inspection.

4.3.1 Titratable iodine determination. Brush twenty tablets with a camel's hair brush to remove adherent powder and dissolve in about 100 mL of distilled water, to which shall be added 1.0 gram of potassium iodide and 2.0 mL of glacial acetic acid. The solution shall be titrated with 0.1 N sodium thiosulfate to a starch endpoint, using exactly 5.0 mL of starch indicator containing 5.0 grams of soluble starch per liter. The equivalent of 1.0 mL of 0.1 N sodium thiosulfate is 12.692 mg of iodine. Divide results by 20 to obtain milligrams of iodine per tablet. To determine titratable iodine in tetraglycine hydroperiodide powder, use 0.8 gram of sample weighed to the nearest 0.1 mg and proceed as above. A blank determination shall be run simultaneously using the same quantities of reagents.

4.3.1.1 Calculation. Calculate mg iodine per tablet as follows:

$$\text{mg iodine/tablet} = \frac{12.692 \times (\text{mLS} - \text{mLB})\ 0.1\text{N}\ Na_2S_2O_3}{20},$$

where mLS = titre for sample and
mLB = titre for blank.

4.3.1.2 Calculation. Calculate percent titratable iodine in tetraglycine hydroperiodide powder as follows:

$$\text{Percent titratable iodine} = \frac{12.692 \times (\text{mLS} - \text{mLB})\ 0.1\text{N}\ Na_2S_2O_3 \times 100}{\text{Weight of sample} \times 1000},$$

where mLS = titre for sample and
mLB = titre for blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

4.3.2 Ash determination. Weigh approximately 2 grams of tetraglycine hydroperiodide to the nearest milligram, place in a platinum crucible with 1 mL of concentrated sulfuric acid. Evaporate the acid and treat again with a few drops of concentrated sulfuric acid. Evaporate and ignite for 20 minutes at a bright red heat. Cool and weigh residue. Calculate percent ash as follows:

$$\text{Percent ash} = \frac{\text{Wt. of ash residue x 100}}{\text{Wt. of sample}}$$

4.3.3 Nitrogen determination. Weigh approximately 0.04 gram of tetraglycine hydroperiodide to the nearest 0.1 mg and transfer to a 125 mL Kjeldahl flask. Add 1.5 grams of a catalyst consisting of 150 parts of K_2SO_4, 5 parts of metallic selenium, and 10 parts mercuric oxide well ground and mixed. Then add 4 mL of concentrated sulfuric acid and digest to a pale yellow color. Digest one-half hour additional, then cool and dilute with 35 mL of distilled water. Transfer the solution to a semimicro Kjeldahl distillation apparatus, add 15 mL of 40 percent NaOH, and mix. Then add 5 mL of 44 percent sodium thiosulfate and distill into 5 mL of 4 percent boric acid solution collecting about 50 mL of distillate. When the distillate gives a negative test for ammonia, disconnect the flask and wash the delivery tube and other apparatus into the distillation flask. Titrate the final distillate with 0.02 N sulfuric acid using a mixed indicator consisting of 10 volumes 0.1 percent bromocresol green solution plus 2 volumes 0.1 percent methyl red solution both in 95 percent alcohol. A blank determination should be made following the exact procedure. Calculate percent total nitrogen as follows:

$$\text{Percent total nitrogen} = \frac{(\text{mLS} - \text{mLB})\ 0.02\text{N}\ H_2SO_4 \times 0.28 \times 100}{\text{Weight of sample (mg)}},$$

where mLS = titration of sample and
mLB = titration of blank.

4.3.4 Moisture content. Weigh approximately 2 to 10 grams of the sample to the nearest milligram in a tared weighing bottle. Place in a drying oven at 120° ± 2°C for 12 hours. Remove, place in a desiccator, and cool to room temperature. Reweigh and calculate moisture content as follows:

$$\text{Moisture content} = \frac{(\text{Wt. of sample before drying} - \text{Wt. of sample after drying}) \times 100}{\text{Wt. of sample before drying}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

4.3.5 Fiber diameter measurement. The measurements shall be made using a suitable air permeability method or microscope with an ocular micrometer (see 4.3.5.1 and 4.3.5.2). When microscopic measurements are made, not less than 100 individual fibers taken at random from the sample shall be measured to determine the average diameter.

4.3.5.1 Microscope calibration. Place the ocular micrometer on the shelf of the 10X eyepiece. Focus on the graduation of the stage micrometer. Use the low power and center the graduation in the field. The stage micrometer has a ruling 2 mm long. At one end of the ruling there are lines 0.01 mm apart. These spacings are the ones to be used for calibration. Turn the eyepiece so that its lines are parallel with those on the stage. Make lines at one end of the ocular and stage micrometer coincide, then look for another line (as far from the first as possible) on the ocular spaces between the lines which coincide at either end.

4.3.5.1.1 Calculation example.

Assume 60 ocular divisions = 9 stage divisions
Since one stage division = 0.01 mm
Then 60 ocular divisions = 0.09 mm
And 1 ocular division = 0.0015

4.3.5.2 Fiber diameter determination. Remove stage micrometer and place an individual glass fiber on the stage and focus on the fiber. Measure the diameter of the fiber by the number of ocular divisions required to span the diameter of the fiber. Multiply the number of ocular divisions required to span the fiber diameter by the millimeter equivalent of one ocular division to convert fiber diameter to millimeters. Convert millimeters to inches by multiplying the thickness of the fiber in millimeters by 0.03937.

4.3.6 Loss on ignition. Weigh approximately 10 grams of an oven-dried sample of glass fiber (dried as specified in 4.3.4) to the nearest milligram in a tared crucible and ignite in an electric muffle furnace. Ignite at 1,200°F for 10 minutes. Upon completion of ignition, allow to cool to room temperature in a desiccator and reweigh. Percent loss on ignition shall be calculated as follows:

$$\text{Percent loss} = \frac{(\text{Wt. dried sample} - \text{Wt. ignited sample}) \times 100}{\text{Wt. dried sample}}$$

4.3.7 Tablet weight. Brush twenty tablets with a camel's hair brush to remove adherent powder, and accurately determine the individual weight of each tablet. The average tablet weight shall be calculated by dividing the sum of

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

the tablet weights by the number of tablets. Determine the individual deviation by subtracting individual tablet weight from the average, or by subtracting the average tablet weight from the individual tablet weight.

4.3.8 Loss on drying. Finely powder 25 tablets, taking care to prevent moisture loss or evaporation. Weigh 2 grams of the powdered tablets to the nearest 0.1 mg, then dry over concentrated sulfuric acid at room temperature for 24 hours. Calculate percent loss on drying as follows:

$$\text{Percent loss on drying} = \frac{(\text{Wt. tablets before drying} - \text{Wt. tablets after drying}) \times 100}{\text{Wt. tablets before drying}}$$

4.3.9 Solubility.

4.3.9.1 After packaging. Place exactly 250 mL of distilled water in a 250 mL glass-stoppered graduated cylinder and adjust the temperature to 20°C. Drop ten tablets into the water, then stopper and invert the cylinder slowly so that the tablets shall be in a continuous travel for 200 seconds. If tablets have not completely dissolved within the time limit specified, immediately after the time period, take a 200 mL aliquot and determine the titratable iodine in accordance with 4.3.1.

4.3.9.2 After 7 days storage at 130°F. When determining the solubility requirements of 3.3.6, the procedure shall be the same as outlined in 4.3.9.1 except that the tablets shall be in continuous travel for 300 seconds. If the tablets have not completely dissolved within the specified time limit, immediately after the time period, take a 200 mL aliquot and determine the titratable iodine in accordance with 4.3.1.

4.3.10 Friability. Accurately weigh 100 tablets free from adhering dust by blowing with a stream of dry air, and place into a 100 mL glass-stoppered graduated cylinder. Stopper the cylinder and continue to invert and re-invert the cylinder until the tablets have traveled the full length of the cylinder 200 times. (This may be facilitated by fastening the cylinder to a wheel that revolves at a rate of not more than 8 revolutions per minute (rpm)). Remove the tablets onto a screen and free them from all pieces of broken tablet and dust by brushing with a camel's hair brush and blowing with a stream of dry air. Count and reweigh only the whole tablets and those showing slight edge erosion. Calculate percent loss of friability as follows:

$$\text{Percent loss} = \frac{(\text{Wt. of 100 tablets} - \text{Wt. of tablets after friability test}) \times 100}{\text{Wt. of 100 tablets before test}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

4.3.11 Wax coating stability. Bottles, packaged as specified, shall be taken from each lot and placed in a temperature not higher than -20°F for a period of not less than 3 hours. Upon examination of each bottle, the wax coating shall exhibit good adhesion to the container and cap closure and shall not show signs of deep cracking or crazing down to the glass or cap surface. Superficial cracking or crazing will not be considered a basis for rejection.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A.

5.1.1.1 Unit pack. The tablets shall be unit packed in quantities of 12 or 50 specified (see 6.2).

5.1.1.1.1 Twelve tablets. Twelve tablets shall be unit packed in a round, amber colored glass bottle conforming to group A, class 1, style 2, grade 1 of PPP-C-186. The bottle shall have overall dimensions as follows: 29.7 ± 0.4 mm in height, 15.5 ± 0.6 mm in diameter, and minimum inside diameter of the mouth of 7.1 mm. Space between bottle closure and tablets shall be filled with a pledget of clear fiberglass as specified in 3.4. Each bottle shall have a primary closure and a secondary moisture vapor-resistant closure as specified in 5.1.1.1.3, except that primary closure shall be closure A of PPP-C-186, and cap shall be provided with vertical ribs around the exterior circumference (see 6.6).

5.1.1.1.2 Fifty tablets. Fifty tablets shall be unit packed in a square or round amber colored glass bottle conforming to group A, class 1, style 2, grade 1 of PPP-C-186. The bottle shall have overall dimensions as follows: 52 ± 1 mm in height, 24.6 ± 1 mm in diameter, and minimum inside diameter of the mouth of 11.1 mm. Space between bottle closure and tablets shall be filled with a pledget of clean fiberglass as specified in 3.4. Each bottle shall have a primary closure and a secondary moisture vapor-resistant closure as specified in 5.1.1.1.3.

5.1.1.1.3 Closure. The primary closure of the clean glass bottle shall be in accordance with closure A or B of PPP-C-186. The interior and exterior of closure B shall be further protected to resist iodine corrosion. The liner for A and B closure shall be faced with a minimum 0.005 inch vinyl film that may be coated, if desired, with an 0.001 to 0.002 inch microcrystalline wax film. The melting point of the wax coating shall be not less than 160°F. Bottle closure shall be effected to obtain a complete and tight seal. A secondary closure shall be obtained by dipping or coating to effect a moisture vapor-resistant seal of cap and bottle juncture with a molten wax consisting of not less than 88 percent microcrystalline type wax combined with not less

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

than 6 percent polyisobutylene. The compound shall have a melting point of not less than 160°F and shall form a clear homogenous composition with no signs of separation when melted and held at 200°F for 1 hour. It shall have a needle penetration of not less than 23 nor more than 28 when tested in accordance with ASTM D5. Final compound shall exhibit good adhesion to the container and cap closure, and shall not show appreciable signs of crazing or cracking when tested as specified in 4.3.11. Secondary closure shall be carried out in such a manner as to effect a neat and complete seal of the bottle and cap juncture that is free from air bubbles. Wax shall cover lower edge of cap and bottle shoulder and not obliterate label marking.

5.1.1.2 Intermediate pack.

5.1.1.2.1 Twelve-tablet bottles. Three hundred 12-tablet bottles, unit packed as specified in 5.1.1.1.1, shall be intermediately packed in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. Each box shall be provided with full height, half-slotted chipboard or paperboard partitions fabricated in such a manner as to provide a snug-fitting cell for each bottle. A fiberboard layer pad shall be provided between each tier or bottle. Twelve instruction sheets specified in 3.5.2 shall be placed in each box. Instruction sheets shall be evenly distributed throughout the box by placing at the top of the box and under each layer pad. Each box shall be closed and placed in a polyethylene bag as specified in 5.1.1.2.2.

5.1.1.2.2 Fifty-tablet bottles. One hundred bottles, unit packed as specified in 5.1.1.1.2, shall be intermediately packed in a snug-fitting fiberboard box conforming to style RSC-L, type CF, variety SW or SF, class domestic, grade 275 of PPP-B-636. The inside of each box shall be fitted with a liner conforming to type CF, variety SW, class domestic, grade 275 of PPP-B-636. Each box shall be provided with full height half-slotted partitions fabricated in such a manner as to provide a snug-fitting cell for each bottle. Each box shall be provided with top and bottom pads and, in addition, multiple tier boxes shall be provided with layer pads. Pads shall conform to type CF, variety SW, class domestic, minimum grade 200 of PPP-B-636. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that metal stitches or staples are not permitted on top and bottom flaps. Each box, with all corners blunted, shall then be placed in a polyethylene bag heat-sealed after all excess air has been expelled. The polyethylene film shall be not less than 0.0015 inch thick conforming to type I, grade A or B, finish 1 or 2 of L-P-378. Seams and closures shall withstand the test specified therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2).

* 5.2.1 Level A packing. Eight intermediate boxes of 12-tablet bottles or twelve intermediate boxes of 50-tablet bottles, preserved as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. When staples are used to form the body joint of the shipping container, the staples shall be covered with pressure sensitive tape. Each container shall be closed, waterproofed, and reinforced with strapping or tape banding in accordance with the appendix of PPP-B-636, except that metal stitches or staples are not permitted on top or bottom flaps and the inspection shall be in accordance with 4.2.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Eight intermediate boxes of 12-tablet bottles or twelve intermediate boxes of 50-tablet bottles, preserved as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, (variety SW) or SF, class domestic, grade 275 of PPP-B-636. When staples are used to form the body joint of the shipping container, the staples shall be covered with pressure sensitive tape. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that metal stitches or staples are not permitted on top and bottom flaps and the inspection shall be in accordance with 4.2.5.

5.2.2.1 Weather-resistant container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.5.

5.3 Palletization. When specified (see 6.2), tablets, packed as specified in 5.2.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, intermediate packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 including the lot number. In addition, contract number and name of contractor shall be shown on the intermediate pack.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

* 5.4.1 Shelf-life marking. Type II shelf-life markings as specified in MIL-STD-129 shall be shown on unit and intermediate packs and exterior containers. The inspection test date shall be 24 months from the date of manufacture and subsequent inspection test dates shall be 24 months thereafter. Each intermediate box of 12-tablet bottles shall be marked with directions for use as specified in 3.5.2, of such size as to be easily read. Marking may be accomplished by affixing the instruction sheet specified in 3.5.2 to the box.

* 5.4.2 Label. A label shall be applied in accordance with MIL-STD-129 to each shipping container and the following legend shall be printed in black ink on red paper:

CAUTION

Not to be used after (see Inspection Test Date) unless tablets yield a minimum of 7.0 mg of titratable iodine per tablet and a maximum solubility time of 300 seconds when tested in accordance with 4.3.1 and 4.3.9 of MIL-W-283H. Lots not meeting above criteria shall be destroyed.

6. NOTES

6.1 Intended use. The water purification iodine tablet is suitable for purification of water in canteens. This item is issued for emergency health protection reasons to provide field troops with drinking water free from disease-bearing organisms.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Whether tablets are to be unit packed in quantities of 12 or 50 (see 5.1.1.1).
c. Selection of applicable level of packing (see 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 Conditions. Compounding, tableting, and packaging should be performed in air conditioned premises at a temperature not more than 80°F, and at a relative humidity of not more than 35 percent at 80°F, and in such a manner as to ensure a product of consistently high quality. The tablets should be packaged in moisture vapor proof containers and maintained under the specified control conditions of temperature and humidity until packaged and sealed in bottles. Tetraglycine hydroperiodide between the time it is

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

manufactured and at the time of compounding should be stored in sealed bottles or equivalent closed containers (see 6.4). It should be moved to air-conditioned quarters when needed and the containers opened only after all the material has come to temperature equilibrium with the surroundings. All other ingredients of the tablets as well as bottles, closures, and pledget material should be kept dry and protected from moisture or high humidity until incorporated into the final package (see 6.5).

6.4 Storage. Avoid storing tetraglycine hydroperiodide at temperature above 90°F.

6.5 Drying. Drying of tablet ingredients (other than tetraglycine hydroperiodide), closures, bottles, and pledget material for 6 hours at 110° ± 2°C has been found satisfactory.

6.6 Caps. Plastic caps with vertical ribs around the exterior circumference, specified in 5.1.1.1.1, are intended to minimize slippage. Caps other than the design specified shall be submitted to the U.S. Army Natick Research, Development, and Engineering Center (Science and Advanced Technology Laboratory) for approval.

* 6.7 Subject term (key word) listing.

Canteen, water
Health protection
Iodine tablet
Purification, water

6.8 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Air Force - 68

Review activities:

Army - MD, EA, ME
Navy - MS
DPSC - DM
DLA - GS

User activities:

Army - AR
Navy - SH, YD, MC

Preparing activity:

Army - GL

Project No. 6850-0832

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

NSN 6850-00-985-7166
WATER PURIFICATION TABLETS,
IODINE 50 FOR TREATING
WATER IN CANTEENS

Active Ingredient
Tetraglycine Hydroperiodide 16.7%
Inert Ingredients 83.3%
Each Tablet Contains 6.68% Titratable Iodine.
Each Tablet Liberates 8.0 mg. Iodine.

Tablets not to be swallowed.

Lot No. E.P.A. Reg. No. 40510-3
MFD INSPECT TEST DATE E.P.A. Est.No.
NAME OF MNFG & ADDRESS

DIRECTIONS FOR USE
IT IS A VIOLATION OF FEDERAL LAW TO USE THIS PRODUCT IN A MANNER INCONSISTENT WITH ITS LABELING.

1. Add two tablets to a one quart canteen of clear water.
2. Replace canteen cap loosely. Wait 5 minutes and shake well, allowing leakage.
3. Tighten cap. Wait an additional thirty minutes before using for any purpose.

Storage and Disposal:

Discard fiber plug. Recap bottle tightly. Keep tablets dry. Do not reuse empty bottle. Discard in a safe place.

PRECAUTIONARY STATEMENTS AND HAZARDOUS TO HUMANS AND DOMESTIC ANIMALS: Tablets may be harmful if swallowed. Avoid eye contact. Will cause severe irritation to eyes and mucous membranes. Avoid contamination of food. Wash after handling. **FIRST AID:** If tablet is swallowed, drink freely a thin paste of starch or flour (2 tablespoons with water) or Sodium thiosulfate (approximately ½ teaspoonful in water.) In case of contact with eyes, flush thoroughly with water. Call a physician. Do not reuse this bottle. Discard in a safe place.

FIGURE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

US Army Natick Research, Development, and
Engineering Center
Natick, MA 01760-5014



NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research, Development,
and Engineering Center
ATTN: STRNC-ES
Box 14A
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-283H	Water Purification Tablet, Iodine

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T06:54Z
Check the source to verify that this is the current version before use.